HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and architected **GPU-accelerated HPC clusters**, including high-speed storage and low-latency networking for enterprise AI workloads.
- Collaborated with sales teams and enterprise customers to translate requirements into scalable **AI/HPC system designs**, covering compute, GPU topology, and interconnect architectures.
- Managed OEM server and appliance configurations, including **Bill of Materials (BOM) definition**, firmware tuning, and manufacturability alignment for complex solutions.
- Delivered technical presentations and demonstrations focused on **AI training and inference workloads**, showcasing system capabilities to potential clients.
- Performed advanced troubleshooting and performance tuning across **compute, GPU, storage, and networking subsystems**, resolving complex system-level issues.
- Developed and maintained technical documentation, including solution architectures and integration guides for **manufacturing and test procedures**.
- Containerized solution components using **Docker** and managed deployments on **AWS EC2 instances**, ensuring efficient resource utilization.

SKILLS

Compute Infrastructure – HPC clusters, AI/ML infrastructure, GPU-accelerated systems, Server architecture (x86/ARM), CPU/GPU platforms (NVIDIA/AMD), Memory, Storage, Networking

High-Speed Interconnects – InfiniBand, RDMA, NVLink, High-performance Ethernet

System Integration & Manufacturing – OEM server platforms, System integration, BIOS/BMC, Firmware, Hardware validation, Rack integration, Burn-in testing, QA, Deployment workflows

Software & Automation – Linux environments, Bash scripting, Python scripting, System automation tools, CUDA, AI frameworks (PyTorch, TensorFlow)

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes

Customer Engagement – Pre-sales architecture, Customer requirements translation, Technical presentations, Demos, Proof-of-Concepts (POCs), Troubleshooting, Performance tuning, Cross-functional collaboration

PROJECTS

AI Training Cluster Design and Validation

- Architected a **multi-node AI training cluster** utilizing NVIDIA GPUs and high-speed InfiniBand interconnects, tailored for deep learning workloads.
- Configured server hardware, including **BOM selection and firmware flashing**, to meet specific performance and thermal requirements for dense compute nodes.
- Developed integration test plans and executed **burn-in testing** across the cluster to validate system stability and performance under heavy load.

HPC Solution Deployment Framework

- Designed a repeatable framework for deploying **HPC infrastructure solutions** on AWS, incorporating OS provisioning and workload manager integration (Slurm).
- Automated cluster bring-up and configuration using **Python scripting** and shell scripts, reducing deployment time by approximately 40%.
- Provided remote support for cluster commissioning and acceptance testing, ensuring successful integration into customer environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)